

Proof of conjecture 00000003419: commute time = $2m \cdot$ effective resistance

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1 The conjecture and its reading

Statement (English, verbatim). Definition: Hitting times: the symmetry of commute times. Conjecture: The commute-time symmetry: the two-point commute time is twice the edge count times the resistance, with the proof of symmetry the symmetric difference of Green functions. (commute resistance twice symmetric difference)

The Chinese text says the same; its Definition line contains the stray word “Holant”, which plays no role in the claim and is ignored.

Reading. $G = (V, E)$ is a finite, connected, simple (unweighted) graph with $m = |E|$ edges. The walk is the simple random walk; every edge is a resistor of one ohm. For vertices a, b the claim is

$$\kappa(a, b) := H(a, b) + H(b, a) = 2m R_{\text{eff}}(a, b),$$

together with the symmetry $\kappa(a, b) = \kappa(b, a)$ and $R_{\text{eff}}(a, b) = R_{\text{eff}}(b, a)$. The clause “the proof of symmetry [is] the symmetric difference of Green functions” describes a proof method. We follow it through the hitting-time vectors $h_b = H(\cdot, b)$: the unit-current potential is the normalized difference $(h_b - h_a)/(2m)$ (Lemma 6). Green functions are not built in Lean as separate objects. The bridge to the Green-function formulation is as follows: with $\Gamma = L^+$ (the Green function of the Laplacian on the orthogonal complement of the constants), $L\Gamma(\cdot, a) = e_a - \frac{1}{n}\mathbf{1}$, so for $a \neq b$ the vector $\Gamma(\cdot, a) - \Gamma(\cdot, b)$ solves $Lv = e_a - e_b$; since the kernel of L is the constants (connectedness), our established equation $L((h_b - h_a)/(2m)) = e_a - e_b$ gives $(h_b - h_a)/(2m) = \Gamma(\cdot, a) - \Gamma(\cdot, b)$ modulo an additive constant. Thus the symmetric difference of Green functions is exactly the potential used in the proof (this remark is not formalized; the main identity does not depend on it).

Conventions. $T_b = \min\{t \geq 0 : X_t = b\}$, so $H(b, b) = 0$. Connectedness is needed, since otherwise hitting times are infinite. The case $a = b$ is allowed, and both sides are then 0. Weighted graphs and multigraphs are not treated.

Attribution. The identity is due to A. K. Chandra, P. Raghavan, W. L. Ruzzo and R. Smolensky, *The electrical resistance of a graph captures its commute and cover times*, first presented at STOC 1989; the journal version, with P. Tiwari as an additional author, appeared in Computational Complexity 6 (1996) 312–340 (Crossref record retrieved: <https://doi.org/10.1007/bf01270385>). Wikipedia’s article “Resistance distance” (https://en.wikipedia.org/wiki/Resistance_distance) defines the resistance distance of a simple connected graph as the resistance “with each edge being replaced by a resistance of one ohm”. It describes the commute time as “the expected number of steps in a random walk that starts at” u , visits v and returns to u , and states $C_{u,v} = 2m\Omega_{u,v}$ with that reference. Our proof is self-contained and fully formalized.

2 Definitions (as in the Lean file)

Let $L = D - A$ be the Laplacian (`Mathlib SimpleGraph.lapMatrix`).

Definition 1. 1. Transition probability: $P(x, y) = 1/\deg x$ if $x \sim y$, else 0 (`trans`). Lemma `sum_trans`: $\sum_y P(x, y) = 1$ when $\deg x > 0$.

2. Trajectory probability of $w = (w_0, \dots, w_t)$: $\prod_{i < t} P(w_i, w_{i+1})$ (`pathProb`). On graphs where every vertex has positive degree this is the law of the Markov chain on the first $t + 1$ steps. On the connected one-vertex graph, `trans` is identically 0, so `pathProb` is not a probability law for $t > 0$; there the only hitting target is the starting vertex, every survival probability is 0, and both sides of the identity vanish, so the theorem still holds (equivalently, one may adopt an absorbing self-loop convention at an isolated vertex).

3. $P_x(T_b > t) = \sum \{\prod_{i < t} P(w_i, w_{i+1}) : w_0 = x, w_i \neq b \text{ for all } i \leq t\}$ (`survProb`).

4. $H(x, b) = E_x[T_b] = \sum_{t \geq 0} P_x(T_b > t)$, the tail-sum formula for the expectation of a random variable with values in $\{0, 1, 2, \dots\} \cup \{\infty\}$ (`hittingTime`, a real `tsum`; the series is proved to converge). $\kappa(a, b) = H(a, b) + H(b, a)$ (`commuteTime`).

5. A unit-current potential from a to b is a $v : V \rightarrow \mathbb{R}$ with $Lv = e_a - e_b$ (`IsUnitPotential`): one unit of current enters at a and leaves at b , and Kirchhoff's and Ohm's laws hold with unit resistors. $R_{\text{eff}}(a, b) = v(a) - v(b)$ (`effRes`, chosen by `Classical.choose`). We prove that such a v exists and that $v(a) - v(b)$ does not depend on the choice. Equivalently, $R_{\text{eff}}(a, b) = (e_a - e_b)^\top L^+(e_a - e_b)$ with L^+ the Moore–Penrose inverse; this remark is not formalized.

3 Proof

Lemma 2 (`survProb_zero`, `survProb_succ`, `survProb_self`). $P_x(T_b > 0) = [x \neq b]$, $P_b(T_b > t) = 0$, and $P_x(T_b > t + 1) = [x \neq b] \sum_y P(x, y) P_y(T_b > t)$.

Proof. Split a trajectory of length $t + 1$ as (w_0, w') using `Fin.consEquiv`. The product factors as $P(w_0, w'_0) \cdot \prod_{i < t} P(w'_i, w'_{i+1})$. Then sum over $w_0 = x$ and group by $w'_0 = y$. $\square$

Lemma 3 (`exists_dirichlet`, `dirichlet_nonneg`). *There is g with $g(b) = 0$, $(Lg)(x) = \deg x$ for $x \neq b$, and $g \geq 0$.*

Proof. Let M be L with row b replaced by $e_b^\top$. If $Mg = 0$, then $g(b) = 0$ and $(Lg)(x) = 0$ for $x \neq b$. Column sums of L vanish (`sum_lapMatrix_mulVec`, from $L^\top = L$ and $L1 = 0$), so $(Lg)(b) = 0$ too. On a connected graph $\ker L$ consists of the constants (`lapMatrix_mulVec_eq_zero_iff_forall_reachable`), so $g = 0$. Thus M is injective, hence invertible and surjective, and we solve $Mg = (\deg x [x \neq b])_x$. Minimum principle: let g attain its minimum at x_0 . If $x_0 \neq b$, then $\deg x_0 > 0$ by connectedness, and $\deg x_0 = (Lg)(x_0) = \sum_{y \sim x_0} (g(x_0) - g(y)) \leq 0$, a contradiction. So the minimum is $g(b) = 0$. $\square$

Lemma 4 (`partial_le`, `summable_survProb`). $\sum_{t < N} P_x(T_b > t) \leq g(x)$ for all N , so $\sum_t P_x(T_b > t)$ converges.

Proof. For $x \neq b$, $(Lg)(x) = \deg x$ means $g(x) = 1 + \sum_y P(x, y)g(y)$ (`lap_eq_degree_iff`). Induct on N . Using the recursion and $P \geq 0$, we get $\sum_{t < N+1} P_x(T_b > t) = 1 + \sum_y P(x, y) \sum_{t < N} P_y(T_b > t) \leq 1 + \sum_y P(x, y)g(y) = g(x)$. The terms are nonnegative, so `summable_of_sum_range_le` applies. $\square$

Lemma 5 (`hittingTime_step`, `lap_hittingTime`). $H(b, b) = 0$, and $H(x, b) = 1 + \sum_y P(x, y)H(y, b)$ for $x \neq b$. Hence $(Lh_b)(x) = \deg x - 2m[x = b]$.

Proof. Shift the convergent series by one and exchange the finite sum with the series. For $x \neq b$ the first-step equation is $(Lh_b)(x) = \deg x$. At b , vanishing column sums and $\sum_x \deg x = 2m$ (`sum_degrees_eq_twice_card_edges`) give $(Lh_b)(b) = -\sum_{x \neq b} \deg x = \deg b - 2m$. $\square$

Lemma 6 (`hitting_potential`). If $a \neq b$, then $m > 0$ and $v = (h_b - h_a)/(2m)$ satisfies $Lv = e_a - e_b$.

Proof. $2m = \sum_x \deg x \geq \deg a > 0$. Also $Lv = ((\deg - 2me_b) - (\deg - 2me_a))/(2m) = e_a - e_b$. $\square$

Theorem 7 (`commute_time_identity`). For every finite connected simple graph G and all vertices a, b : both tail series converge; a unit-current potential exists; for $a \neq b$, $(h_b - h_a)/(2m)$ is one; every unit-current potential v gives $\kappa(a, b) = 2m(v(a) - v(b))$; $\kappa(a, b) = 2m R_{\text{eff}}(a, b)$; $\kappa(a, b) = \kappa(b, a)$; and $R_{\text{eff}}(a, b) = R_{\text{eff}}(b, a)$.

Proof. If $a = b$, both sides are 0. Otherwise, two unit potentials differ by an element of $\ker L$, i.e. by a constant (`potential_diff_eq`). So for any unit potential v , $v(a) - v(b) = [(h_b(a) - h_a(a)) - (h_b(b) - h_a(b))]/(2m) = (H(a, b) + H(b, a))/(2m)$, using $H(a, a) = H(b, b) = 0$. The same holds for R_{eff} (`effRes_eq`). κ is symmetric by definition. R_{eff} is symmetric because $-v$ is a unit potential from b to a (`effRes_comm`). $\square$

4 Lean formalization and axiom audit

The file `lean/Conjecture3419/Basic.lean` (315 lines, `import Mathlib` only) contains all the definitions and lemmas above, in namespace `C3419`. The main theorem is as follows (Lean's conjunction, existential, inequality, implication and universal quantifier are written here in ASCII as `&`, `exists`, `!`, `->` and `forall`):

```
theorem commute_time_identity (hG : G.Connected) (a b : V) :
  (Summable fun t => survProb G b t a) /\
  (Summable fun t => survProb G a t b) /\
  (exists v, IsUnitPotential G a b v) /\
  (a != b -> IsUnitPotential G a b fun x =>
    (hittingTime G x b - hittingTime G x a) / (2 * #G.edgeFinset)) /\
  (forall v, IsUnitPotential G a b v ->
    commuteTime G a b = 2 * #G.edgeFinset * (v a - v b)) /\
  commuteTime G a b = 2 * #G.edgeFinset * effRes G a b /\
  commuteTime G a b = commuteTime G b a /\ effRes G a b = effRes G b a
```

Here V is a `Fintype` with decidable equality and $G : \text{SimpleGraph } V$. The command `lake build` succeeds. `#print axioms C3419.commute_time_identity` reports only `[propext, Classical.choice, Quot.sound]`. There is no `sorry` and no `native_decide`.